Automated Cucumber Disease Recognition Using Convolutional Neural Networks

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***Abstract*—Cucumber diseases reduce crop yield and quality, making early and accurate diagnosis essential. This paper presents an automated cucumber disease recognition system using Convolutional Neural Networks (CNNs) to classify disease types from leaf and fruit images. The proposed pipeline applies preprocessing and augmentation, and compares multiple CNN architectures based on predictive performance and computational efficiency (training time). Results indicate that deep learning-based methods can support farmers and agricultural experts with rapid disease detection.**

***Index Terms*—Cucumber disease recognition, convolutional neural networks, transfer learning, CBAM, image classification**

I. Introduction

Cucumber is an important vegetable crop, but it is vulnerable to various diseases that can significantly reduce yield and quality. Conventional disease identification is typically performed manually by farmers or agricultural experts, which is time-consuming, subjective, and often unavailable in resource-limited settings. With recent progress in computer vision and deep learning, Convolutional Neural Networks (CNNs) have become effective for automated plant disease recognition by learning visual patterns such as color changes, texture, and lesion structures from images.

This paper investigates automated cucumber disease recognition using CNN-based models and evaluates different architectures in terms of classification accuracy and training efficiency.

II. Related Work

Recent research on cucumber disease recognition has primarily focused on supervised deep learning approaches using convolutional neural networks. Transfer learning with architectures such as VGG, ResNet, and EfficientNet has consistently shown strong performance, while recent studies have explored hybrid and attention-based models to further enhance accuracy and robustness.

Compared to prior work, the present study evaluates multiple pretrained CNNs, a custom CNN, and an attention-enhanced CBAM-ResNet50 using a unified dataset and evaluation protocol. In addition to classification accuracy, computational efficiency and practical deployability are emphasized, addressing key limitations identified in existing literature.

III. Methodology

A. Dataset and Classes

The cucumber condition dataset consists of eight classes representing diseased and healthy samples. Six classes correspond to diseases: Anthracnose, Bacterial Wilt, Belly Rot, Downy Mildew, Pythium Fruit Rot, and Gummy Stem Blight. Two classes represent healthy conditions: Fresh leaves (healthy leaf) and Fresh cucumber (healthy fruit). The dataset includes 1,280 original images in total, with 160 images per class (8 classes $\times$ 160 images/class = 1,280 images).

B. Data Preparation and Split

All images were resized to a fixed input resolution suitable for CNN-based classification. A consistent train/validation/test split was applied across all experiments to ensure fair comparison among models. The same test set was used for every architecture evaluated.

C. Preprocessing and Augmentation

To improve generalization and reduce overfitting, preprocessing steps such as normalization were applied. Data augmentation techniques were used during training to increase effective data diversity, including random rotations, horizontal/vertical flips, and brightness variations.

D. Model Architectures Evaluated

1) Fine-Tuned Pretrained Models: Transfer learning was performed using pretrained CNN backbones trained on ImageNet, including ResNet50, VGG16, and EfficientNetB0. For each model, feature extraction was carried out using the pretrained base, followed by training a custom classification head for the eight-class classification task.

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TABLE I

Comparison of Existing Cucumber Disease Recognition Studies

Study	Dataset Description	Methodology	Accuracy	Key Contributions / Limitations
Raufer *et al.* (2025)	1,280 RGB images, 8 classes, collected under natural conditions in Bangladesh	Modified VGG19 with novel transfer learning	95.31%	High generalization with explainability; limited to a single geographic region
Silva and Almeida (2024)	15,444 thermal images across 7 disease classes	MobileNet variants with pruning and quantization-aware training	Real-time inference	Edge deployment capability; sensitive to environmental temperature
Singh and Kumar (2023)	3,700 cucumber leaf images (augmented to 11,100)	ResNet50-based deep CNN with data augmentation	98.0%	High accuracy using transfer learning; risk of overfitting with large models
Ozguven (2020)	175 high-resolution cucumber leaf images	Faster R-CNN for disease detection and severity estimation	94.86%	Severity-aware detection; small and non-public dataset
Omer *et al.* (2022)	8,000+ images, 6 disease classes	Hybrid VGG16 + CNN + LSTM with IMFO optimization	99.69%	Very high accuracy; computationally expensive model

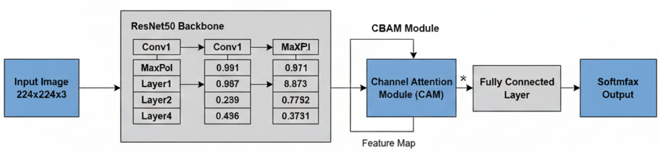

Fig. 1. Architecture of the proposed CBAM-enhanced ResNet50 model for cucumber disease recognition.

2) Custom CNN: A lightweight custom CNN architecture was designed and trained from scratch without transfer learning. This model learns features directly from the cucumber dataset and serves as a baseline to compare against pretrained approaches.

3) Attention-Enhanced CNN (CBAM): An attention-enhanced architecture was implemented by integrating the Convolutional Block Attention Module (CBAM) into the pretrained ResNet50 backbone. CBAM refines feature representation by emphasizing informative channels and spatial regions, allowing the network to focus on critical disease patterns.

E. Evaluation Metrics

The performance of all models was evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC/AUC.

TABLE II

Performance Summary of Evaluated Models

Model	Accuracy	Training Time (min)
ResNet50	0.933	96.27
VGG16	0.897	203.51
EfficientNetB0	0.830	84.31
Custom CNN	0.773	12.72
CBAM-ResNet50	0.890	197.54

IV. Experiments and Results

A. Performance Summary

Top Performer: The Fine-Tuned ResNet50 model achieved the highest predictive performance with an outstanding 93.3 percentage Accuracy and 0.93 Weighted F1-score. This confirms that the rich, general-purpose features learned by ResNet50 on the massive ImageNet dataset are highly transferable and effective for this specific disease recognition task. Custom Model Viability: Custom CNN achieved a respectable 77.32percentage Accuracy, proving that a scratch-built model can perform adequately, though it lags behind all of the successfully fine-tuned pre-trained models. Efficiency Analysis Most Efficient: Custom CNN is the most computationally efficient model, completing its training in a mere 12.72 minutes. This makes it an ideal choice for environments with severe computational resource constraints. Least Efficient: The VGG16 model was the slowest, requiring over three hours (203.51 minutes) for training, demonstrating the inefficiency of its older, deep architecture.

V. Discussion

The results highlight that transfer learning is effective for cucumber disease recognition, especially with limited dataset size. Attention mechanisms such as CBAM can refine feature focus, but may increase computational cost depending on implementation and training settings. Future work may include collecting more field-condition images, optimizing attention integration for faster training, and deploying the best-performing model on mobile/edge devices for real-time decision support.

VI. Conclusion

The evaluation establishes a clear trade-off between performance and efficiency across the models: Recommendation for Accuracy: The Fine-Tuned ResNet50 is the recommended model for deployment, offering the highest reliability (93.3 percentage Accuracy). Its superior predictive capability is essential where misclassification of diseases carries a high cost. Recommendation for Efficiency: Custom CNN is the superior choice when training time and resource consumption are the primary constraints, providing a fast solution (12.72 min) with acceptable performance (77.32The attention mechanism, as implemented in this final task, failed to provide a performance boost, indicating that for this dataset, simple fine-tuning of robust pre-trained models is the most effective approach.

This paper presented an automated CNN-based framework for cucumber disease recognition. Experimental results confirm the effectiveness of pretrained and attention-enhanced architectures for accurate disease classification.

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